2026 MCM Junior Group Problem
The Burger Battle

There are two restaurants, BunBop and PattyPop, selling the same type of burgers near a university. Every morning, each restaurant may buy freshly made burgers from two suppliers and choose a selling price. The number of customers changes from day to day. Burgers that remain unsold at the end of the day must be thrown away. Each restaurant wants to earn as much money as possible, but buying too many burgers may lead to losses. Buying too few may mean losing customers. How should the two restaurants make their decisions?

To simplify the setting, we introduce the following fictional game. All money is measured in **coins**.

Opening. Each restaurant starts with **30 coins and no burgers**. Each day it chooses to open or close for the whole day. Opening costs **4 coins** to cover expenses such as labor costs, paid before buying burgers from suppliers or making sales. No borrowing is allowed. A shop with fewer than 4 coins must close: it cannot pay the opening cost. A closed shop keeps its cash but cannot reopen for that day. Cash carries over between days. The shops cannot transfer money or inventory to each other.

Purchasing decisions. Both restaurants can buy from two suppliers, LittleFlame and MegaGrill. Their burgers are of the same quality, but Supplier MegaGrill charges more because it is farther away.

Supplier	Price per burger	Daily purchasing limits for each restaurant
LittleFlame	2 coins	4 burgers
MegaGrill	4 coins	8 burgers

The purchasing limits are separate and reset each day. Unused purchasing limits cannot be shared or saved. Thus, an open restaurant can buy at most 12 burgers, provided it has enough cash. Since the burgers have the same quality, it is cheaper to buy from Supplier LittleFlame first and then Supplier MegaGrill.

The opening cost plus the cost of purchased burgers cannot exceed the restaurant's cash at the start of the day. For simplicity, we assume that it cannot buy more burgers later that day.

Selling prices. Each restaurant chooses one price: an integer from 1 to 8 coins per burger. Purchasing quantities and selling prices are chosen privately before customers arrive and cannot be changed during the day. Both restaurants know all earlier days' choices and results of these two restaurants, but learn each other's choices for the current day only after the day ends. Thus, they cannot use the other restaurant's current choices when making their decisions.

Customers. The number of customers depends on the day. There are three types of days, each occurring with equal probability, independently from day to day. The restaurants do not know the day's type before making their choices.

Table 1: Type of Days

Type of day	Number of Customers	Highest price each customer will pay
Quiet day	6	4 coins
Normal day	12	6 coins
Festival day	18	8 coins

Each customer wants at most one burger. Customers buy one at a time from the cheaper restaurant that still has burgers, provided its price does not exceed the highest price in Table 1. If neither restaurant qualifies, they leave without buying. When both restaurants charge the same price, each customer chooses one at random. If that restaurant sells out, the customer may buy from the other one if it still has burgers at an affordable price. Unmet demand does not carry over to the next day.

Closing time. Unsold burgers are thrown away and have no value. For an open shop,

$$\text{new cash amount} = \text{old cash amount} - 4 - \text{purchasing cost} + \text{selling price} \times \text{burgers sold}.$$

A closed shop's cash stays unchanged.

A small example. Both shops start with 30 coins, buy 4 burgers each, and charge 4 coins per burger. On a quiet day, suppose each sells 3 burgers and throws away one. Each ends with $30 - 4 - 4 \times 2 + 3 \times 4 = 30$ coins. They sold burgers but made no profit!

Questions

(1) **Start with one day.** Suppose the shops compete with each other for one day to maximize their own final cash. If you run one of the restaurants, what is your best strategy, including whether to open, how many burgers to buy, and what selling price to set? Recall that we do not know the type of today when making decisions. Both shops start with 30 coins.

Hint: Compare at least two plans for the number of burgers purchased and the selling price. For each plan, calculate your restaurant's gain or loss on a quiet, normal, and festival day. The three day types are equally likely. Can you find a best one-day plan? Explain your reasoning or how you searched.

(2) **Now run for 30 days.** Under the same assumptions, design a strategy for running a restaurant for 30 days. Does a good one-day plan remain a good idea when the running time is longer?

Hint: A simple and cautious strategy is as follows. Buy 4 burgers and charge 4 coins per burger every day. If it cannot afford 4 burgers after paying the opening fee, it buys as many as it can, up to 4; if it cannot afford the opening fee, it closes.

Can your strategy do well against different rivals, rather than only one weak opponent? Compare the money earned and the number of burgers sold. Do lower prices always help?

(3) Suppose you are a student at this university. Will you always benefit from the battle between these two shops?

Instructions

- **Use of AI Tools**: Teams are allowed to use AI tools, including ChatGPT, DeepSeek, and other AI-assisted software. However, all AI use must be clearly disclosed. In the report, please include a brief paragraph on *Use of AI Tools* at the end of your submission, including the tools used and their purposes. All AI-generated content, calculations, code, and references must be carefully checked for correctness and plagiarism. Teams remain fully responsible for their submitted work.
- **Submission**: Each team should submit a short report (**no more than 10 A4 pages, font size no smaller than 11pt**) in English explaining your answers and ideas. Please include the most important figures, tables, mathematical models, and numerical results in the report.

  Please list the names of all members in your report. Each team only has to ask the Team Leader to submit the files through the CMT system before October 20, 2026, 23:59 Beijing Time:

  `https://cmt3.research.microsoft.com/HKUSTGZMCM2026`

  - Main submission file: The report, in doc, docx, or pdf format.
  - If there are any source codes used, please submit them as **Supplementary Materials** in a **ZIP** file no larger than 50 MB.
- **Copyright:** Each team retains the copyright of their work. We will only use your submissions to decide the winner.
- **Academic Integrity:** This competition, like all other competitions, functions best when the participants treat one another with honesty, fairness, respect, and trust. The organizers hold all participants of this competition to high standards of scholarship and integrity.

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